

Unified compact model for the ballistic quantum wire and quantum well metal-oxide-semiconductor field-effect-transistor

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We present a compact model based on the Landauer transmission theory for the silicon quantum wire and quantum well metal-oxide-semiconductor field effect transistor (MOSFET) working in the ballistic limit. This model captures the static current-voltage characteristics in all the operation regimes, below and above threshold voltage. The model provides a basic framework to account for the electronic transport in MOSFETs, being easily adaptable to gate structures such as the double-gate or gate-all-around. Numerical simulations based on the proposed model have been compared with experiments and quantum mechanical self-consistent simulations, with good agreement. © 2003 American Institute of Physics. [DOI: 10.1063/1.1582557]

I. INTRODUCTION

The control of short-channel effects is more challenging with the progressive scaling of metal-oxide-semiconductor field-effect-transistors (MOSFETs). The double-gate (DG) and gate-all-around (GAA) MOSFET structures have emerged as main candidates to provide the electrostatic integrity needed to scale down MOSFETs to minimal channel lengths. In addition to a better electrostatics than the single-gate (SG) MOSFET, the use of these devices has advantages relative to the carrier transport, mainly due to (i) reduced surface roughness scattering due to the lower vertical electric field and (ii) the reduction of the Coulomb scattering because the channel is made of undoped/low-doped silicon. Moreover, the introduction of new materials to enhance the mobility, as the strained silicon, can result in electronic transport approaching the ballistic regime. The International Roadmap for Semiconductors (ITRS-200) has identified a series of challenges that must be solved before these devices can enter the manufacturing of MOSFET-based integrated circuits.¹ Among them, a complete compact modeling of these structures remains an issue. In an attempt to contribute to this field, we propose a physical compact model of the ballistic DG and GAA MOSFET operation, based on the Landauer transmission theory. We introduce the extension of existing models, only formulated for quantum well MOSFETs, to

structures of lower dimensionality, such as the quantum wire MOSFET. The Landauer theory has proved to be an excellent framework for describing mesoscopic transport.^{2–7} In this approach, the current through a conductor is expressed in terms of the probability that an electron can transmit through it. The net current of the ballistic MOSFET is obtained by subtracting the drain to source transmitted current from the source to drain transmitted current. The electrons injected from the two electrodes see a potential barrier along the transport direction. We will consider a classical potential barrier with a shape governed by the gate and drain voltages. The transmission probabilities for the electrons crossing this barrier is one or zero depending on whether the electron energy is above or below the barrier, respectively. The use of a classical barrier to model MOSFET operation is a simplification, but it captures the basic mechanism controlling the current. This approach is valid as long as the source to drain tunneling current does not dominate the total current. More detailed current-voltage ($I-V$) characteristics could be obtained considering a quantum potential barrier.

We will generate a compact model for the n -type quantum wire MOSFET shown in Fig. 1(a). In quantum well MOSFETs the electron gas can freely move in two dimensions, along the channel length (x -direction) and channel width (y -direction), though the movement on the channel depth direction (z) is restricted to a finite set of k_z ($\mathbf{k}$ is the wave vector) [Fig. 1(a)]. The number of allowed k_z states are

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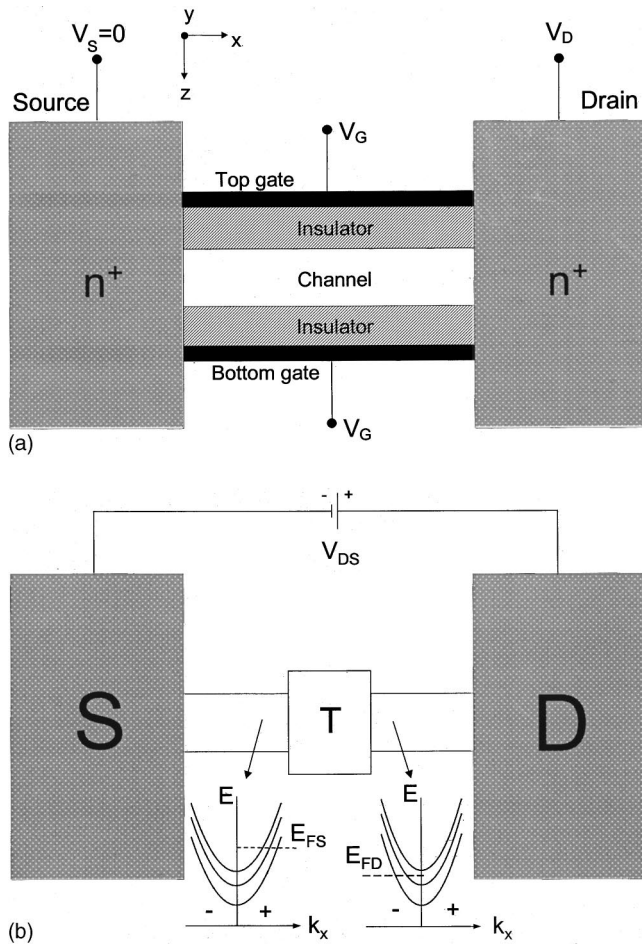


FIG. 1. (a) Cross section of a quantum wire MOSFET, where y and z are the confinement dimensions. The electronic transport is along the x direction; (b) conceptual picture of the electronic transport in a quantum wire. Electrons traveling towards the drain (source) are distributed in discrete subbands populated up to E_{FS} (E_{FD}). The injected electrons from one contact have a transmission probability $T(E)$ to reach the opposite contact.

the number of modes (or subbands) below the Fermi level. In quantum wire MOSFETs the electron gas is also restricted in the y -direction resulting in a finite set of k_y . If the channel width (W) is gradually increased, the current increases in discrete steps due to new contributing modes. In the limit case of very large width, the proposed model recovers the physics of the quantum well MOSFET. In this sense, it provides a unified framework to both the quantum wire and quantum well MOSFETs.

In Sec. II, where the electronic transport is addressed, the model is common to the SG and DG structures. However, the electrostatic behavior has to be considered separately. In Sec. III, we address the electrostatics of the DG structure, that presents a higher potential for the continuous progress in digital^{8,9} and analog applications.^{10,11} In Sec. IV, we show an example to illustrate the prediction capacity of the compact model to capture the current-voltage characteristics of a ballistic quantum wire DG-MOSFET. The proposed model also works for very large width devices, where the electron gas confinement in the y -direction has been removed. To demonstrate this point we compare the predicted current-voltage characteristics of a quantum well DG-MOSFET with self-

consistent simulations (Sec. V). In Sec. VI, we extend the compact model of the DG structure to a cylindrical GAA structure, which offers an excellent control of short-channel effects. For this geometry we can compare with experimental results taken from the literature. Finally, in Sec. VII, we present the main conclusions.

II. MODELING ELECTRONIC TRANSPORT

The conceptual picture of a generic quantum wire MOSFET is that of a conductor connected to two large contacts (source and drain regions) by two leads [Fig. 1(b)]. The contacts are supposed to be in thermodynamic equilibrium so that each can be described by its own Fermi level. The leads are assumed to be ballistic conductors. T is the average probability that an electron injected in lead S will transmit to lead D . In a classical approach, this transmission probability is one or zero whether the electron energy is above or below the potential barrier. We assume that electrons can exit from the conductor into the contacts without any reflections; that is, the contacts are supposed to be perfectly absorbing. Then the $+k_x$ states in lead S are occupied only by electrons coming in from the source and hence these states must have a Fermi level E_{FS} . Similarly the $-k_x$ states must have a Fermi level E_{FD} . The gate electrode (not shown) controls the charge at the beginning of the channel. In equilibrium ($V_{DS} = 0$), a symmetrical potential barrier along the channel length appears because of the built-in potential at the source (n^+)-channel (i) and channel (i)-drain (n^+) junctions, respectively. The Fermi level of the drain and source contacts are equal and no net current flows through the channel, because forward traveling electrons ($+k_x$) are exactly compensated by the backward traveling electrons ($-k_x$). When a positive voltage V_{DS} is applied between drain and source contacts, the Fermi level at the drain contact is reduced qV_{DS} with respect to the Fermi level at the source; i.e., $E_{FD} = E_{FS} - qV_{DS}$. In this situation no net current exists for electrons with energies below E_{FD} , because forward electrons are exactly compensated with backward electrons. However, a net flow of forward electrons, with energies lying between E_{FS} and E_{FD} , appear because energies above the Fermi level corresponding to the backward electrons are unpopulated (at zero Kelvin operation temperature). The effect of an applied voltage over the gate and drain electrodes on the potential barrier shape is illustrated in Fig. 2(a) (the Fermi level is taken as a reference). The separation between the Fermi level and the maximum of the barrier increases with the applied gate-source voltage (V_{GS}), and it will be determined in the next section. The effect of V_{DS} is increasing the potential barrier at the lead D . In Fig. 2(b) the typical energy band diagram along the z -direction is shown, indicating the surface (ψ_s) and center potentials (ψ_0), and the intrinsic Fermi level (E_{Fi}). The current through the device can be obtained by means of the Landauer formula¹²

$$I = \frac{q}{L} \sum_v \sum_{k_z, k_y} \sum_{k_x > 0} g_v [f_S(E) - f_D(E)] \left(\frac{\hbar k_x}{m_x^v} \right) T(E). \quad (1)$$

In this equation, L is the channel length, corresponding to the transport direction (x -direction), and $\hbar k_x / m_x^v$ is the velocity

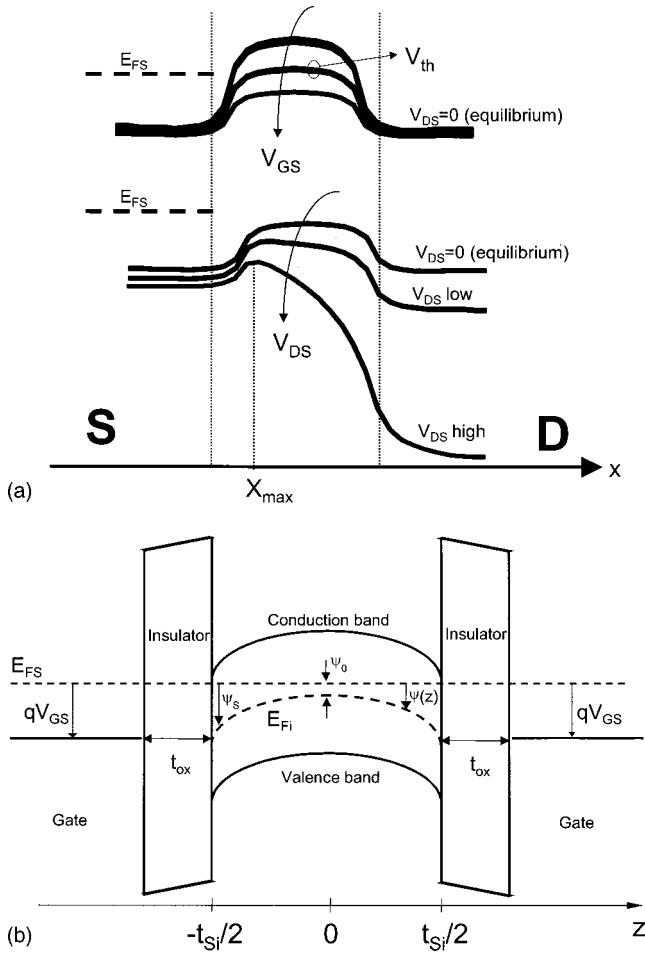


FIG. 2. (a) Profile of the first subband energy along the x direction. The maximum energy point at $x_{\max}$ is at equilibrium. The charge at this point, which represents the virtual cathode, is essentially controlled by the gate electrode. The gate voltage reduces the maximum energy of the barrier. The alignment of the Fermi level with the maximum energy occurs at the threshold voltage (V_{th}). The drain voltage increases the barrier that feel the electrons injected from the drain; (b) energy band diagram along the z -direction.

of electrons as they are injected from the contact with energy E . The two inner sums account for all the wave vectors injected by the electrodes and the outer sum for the three sets of silicon valleys ($v=1,2,3$) where electrons can lie, each one with a two-fold degeneracy ($g_v=2$); m_x^v is the effective mass for the transport in the x -direction. If we suppose that the silicon-oxide interface orientation is $\langle 100 \rangle$, then $m_x^1 = m_x^2 = m_T$, $m_x^3 = m_L$, where $m_T = 0.19m_0$ and $m_L = 0.91m_0$, being m_0 the free electron mass. The electron energy is related with $\mathbf{k}$ through the parabolic dispersion relation; $f_S(E)$ in Eq. (1) is the Fermi function for the source contact, which gives the probability that such an electron is injected from the contact; $T(E)$ is the transmission coefficient for the electron; and $f_D(E) = f_S(E + qV_{DS})$ is the Fermi function at the drain contact. The electron energy E can be written as the sum of the component in the transport direction (E_x) and the component in the confining zy -plane (E_n^v), with the subindex n representing discrete subbands ($E = E_x + E_n^v$); $n=1,2,3,\dots$, and the subindex v representing the valley. Only the E_x component is continuous and can extend from zero to infinity. Assuming periodic boundary

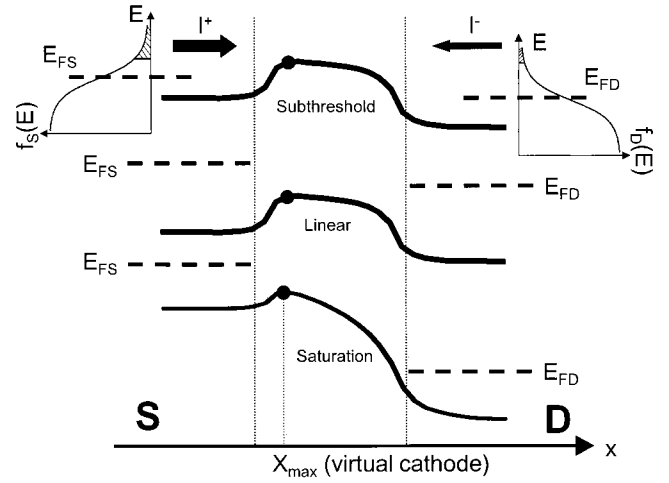


FIG. 3. Energy band diagram for the three possible operation regimes of the MOSFET.

conditions over the x -direction, the inner sum of Eq. (1) can be calculated integrating over k_x and multiplying by the one-dimensional density of states $2x(L/2\pi)$, where the factor 2 accounts for the electron spin. Then, the current can be finally expressed as

$$I = \frac{q}{\pi\hbar} \sum_v \sum_n g_v \left\{ \int_{E_x} [f_S(E) - f_D(E)] T(E) dE_x \right\}, \quad (2)$$

where we have made use of the parabolic dispersion relation for the E_x component

$$E_x = \frac{\hbar^2 k_x^2}{2m_x^v}. \quad (3)$$

The quantity $q/\pi\hbar$ is the current carried per occupied subband per unit energy, which is about 80 nA/meV. For ballistic transport $T(E)=1$ for $E > E_n^v(x_{\max})$, where $x_{\max}$ identifies the position of the n th-subband maximum energy [Fig. 2(a)]. We assume $T(E)=0$ otherwise. Eq. (2) can be developed obtaining the following result:

$$I = \frac{qkT}{\pi\hbar} \sum_v \sum_n g_v \ln \left\{ \frac{1 + e^{[E_{FS} - E_n^v(x_{\max})]/kT}}{1 + e^{[E_{FD} - E_n^v(x_{\max})]/kT}} \right\}. \quad (4)$$

At zero Kelvin temperature, Eq. (4) reduces to

$$I = \frac{q}{\pi\hbar} \sum_v \sum_n g_v \{ H[E_{FS} - E_n^v(x_{\max})][E_{FS} - E_n^v(x_{\max})] - H[E_{FD} - E_n^v(x_{\max})][E_{FD} - E_n^v(x_{\max})] \}, \quad (5)$$

where $H(E)$ is the Heaviside function [$H(E)=1$ if $E>0$; $H(E)=0$ otherwise]. As we will see in the next section, $E_{FS} - E_n^v(x_{\max})$ depends on V_{GS} and V_{DS} . Equation (5) reveals three possible operation regimes depending on the location of the subbands with respect to the Fermi levels (Fig. 3). To discuss these regimes, we define $E_n(x_{\max})$ as the ordered set of energies $E_n^v(x_{\max})$, being $E_1(x_{\max})$ the lowest energy in this set.

(a) $E_{FS} \leq E_1(x_{\max})$: At zero Kelvin the current is zero because the population of $+k_x$ ($-k_x$) states at source and drain contacts have not enough energy to transmit across the barrier towards the drain (source). If the temperature is greater than zero, the tails of the Fermi functions extend above the barrier. Moreover, if the device is out of the equilibrium ($V_{DS} > 0$) some leakage current flows because the tail of $f_S(E)$ above the maximum of the barrier, which is transmitted towards the drain, is larger than the corresponding tail of $f_D(E)$ that cross the channel ballistically towards the source. This leakage current corresponds to the diffusion current. This regime takes place when $V_{GS} < V_{th}$, where V_{th} (threshold voltage) is the gate voltage for the first subband to cross the Fermi level at the source; i.e., the gate voltage for which $E_1(x_{\max}) = E_{FS}$. The situation described corresponds to the subthreshold regime.

(b) $E_{FS} > E_i(x_{\max})$, where $i \geq 1$; and $E_{FD} > E_j(x_{\max})$, where $j \geq 1$: The forward I^+ (backward I^-) current is given by the electrons with energies between $E_i(x_{\max})$ [$E_j(x_{\max})$] and E_{FS} (E_{FD}) (at zero Kelvin). The net current is the difference $I^+ - I^-$, and the slope (G) of the output characteristic is an even multiple of the quantum conductance (G_0): $G = dI/dV_{DS} = 2MG_0$, where $G_0 = q^2/\pi\hbar$; the factor 2 arises from the two-fold degeneracy of each set of valleys and M is the number of subbands below E_{FD} . This situation takes place when $V_{GS} > V_{th}$ and V_{DS} is low enough. It corresponds to the linear regime of the MOSFET operation.

(c) $E_{FS} > E_i(x_{\max})$, where $i \geq 1$; and $E_{FD} \leq E_1(x_{\max})$: In this case, the forward current I^+ is given by the electrons with energies between $E_i(x_{\max})$ and E_{FS} . The backward current I^- is suppressed because the occupied $-k_x$ states at the drain contact cannot transmit across the barrier towards the source. This case corresponds to the saturation regime ($V_{GS} > V_{th}$ and V_{DS} high), for which the current is independent of the applied source-drain voltage. The onset of the saturation regime occurs at the drain voltage $V_{DS,sat}$ that makes $E_1(x_{\max}) = E_{FD}$.

III. MODELING ELECTROSTATICS OF DG-MOSFET

In the last section we have derived the current equation of the quantum wire MOSFET in terms of the separation between the Fermi level and the bottom of the subbands at the maximum energy point ($x_{\max}$). The goal of this section is to find how this separation depends on the applied gate and drain voltages for the DG structure. We will assume that the two-dimensional short-channel effects are small (future extensions of the compact model will deal with two-dimensional electrostatics).

The potential and charge distribution of the intrinsic DG structure has been reported by Taur.^{13,14} The electrostatic analysis was only done in the vertical direction, but it serves our purposes if we apply the results at the maximum energy point ($x_{\max}$). At this special point, the potential and charge distributions are essentially controlled by the electric field perpendicular to the silicon-insulator interface; the lateral electric field is close to zero, and the gradual channel approximation applies.¹⁵ The potential along the vertical direction ($0 \leq z \leq t_{Si}/2$) is given by^{13,14}

$$\frac{q(\psi - \psi_0)}{2kT} = -\ln \left[\cos \left(\sqrt{\frac{q^2 n_i}{2\epsilon_{Si} kT}} e^{q\psi_0/2kT} z \right) \right], \quad (6)$$

where $\psi_0 \equiv \psi(z=0)$ is the potential at the center of the silicon film, ϵ_{Si} is the permittivity of silicon and n_i is the intrinsic carrier density (see Appendix A for a detailed derivation of this quantity in a quantum wire). The surface potential is $\psi_s \equiv \psi(z=t_{Si}/2)$; ψ_s is also related to V_{GS} and t_{ox} through the boundary condition at the silicon-oxide interface^{13,14}

$$\epsilon_{ox} \frac{V_{GS} - \Delta\phi - \psi_s}{t_{ox}} = \epsilon_{Si} \frac{d\psi}{dz} \Big|_{z=t_{Si}/2} = Q/2 = \sqrt{2\epsilon_{Si} kT n_i (e^{q\psi_s/kT} - e^{q\psi_0/kT})}. \quad (7)$$

Here, Q is the sheet density of mobile charge and $\Delta\phi$ is the work-function difference between the gate electrode and intrinsic silicon

$$\Delta\phi = \phi_m - (\chi_{Si} + E_g/2q), \quad (8)$$

where E_g is the gap energy for silicon (1.12 eV). Below the threshold voltage $V_{th} \approx E_g/2q + \Delta\phi$, the mobile charge is very small and $\psi_s \approx \psi_0 \approx V_{GS} - \Delta\phi$. In this case, the bands move as a whole because ψ_s and ψ_0 closely follow V_{GS} . When the gate voltage is further increased above the threshold voltage, when $\psi_s \approx E_g/2q$, the right-hand side (RHS) of Eqs. (6) and (7) are no longer negligible. Since the angle of the cosine function in Eq. (6) cannot exceed $\pi/2$, ψ_0 saturates to the maximum value $(kT/q) \ln(2\pi^2 \epsilon_{Si} kT / q^2 n_i t_{Si}^2)$. On the other hand, ψ_s continues to increase slowly as governed by Eq. (7) with the $\exp(q\psi_0/kT)$ term in the square root neglected [see for example the surface and center potentials in Fig. 4(c)]. From Eq. (7), the sheet density of mobile charge can be written as

$$Q = 2C_{ox}(V_{GS} - \Delta\phi - \psi_s), \quad (9)$$

where ψ_s is a function of V_{GS} that can be solved iteratively from Eq. (7). The gate electrodes control the mobile charge (Q) at the virtual cathode ($x_{\max}$). This charge is provided by the source (Q^+) and drain (Q^-) reservoirs, by adjusting adequately the separation between E_{FS} and $E_n^v(x_{\max})$ ^{2,6,7}

$$Q = \sum_v \sum_n \{Q^+[E_{FS} - E_n^v(x_{\max})] + Q^-[E_{FD} - E_n^v(x_{\max})]\}. \quad (10)$$

This approach assumes that the tunnel current from the gate is negligible. In equilibrium ($V_{DS} = 0$), the k_x -distribution at $x_{\max}$ is symmetrical ($+k_x$ and $-k_x$ states are equally populated). When V_{DS} increases, the negative part of the k_x -distribution is progressively reduced because the barrier height for the $-k_x$ states increases with V_{DS} . To maintain Q constant, the Fermi level at the source must be pulled up to inject more forward traveling electrons into the channel to compensate the reduction of the negative flow. Equation (10) then relates V_{GS} and V_{DS} , with the distance between the Fermi level and the bottom of the subbands at the virtual cathode. It can be written in the form

$$Q = \frac{q}{W} \sum_v \sum_n g_v \int_0^\infty \frac{N_{1D}^v(E)}{2} [f_S(E) + f_D(E)] dE. \quad (11)$$

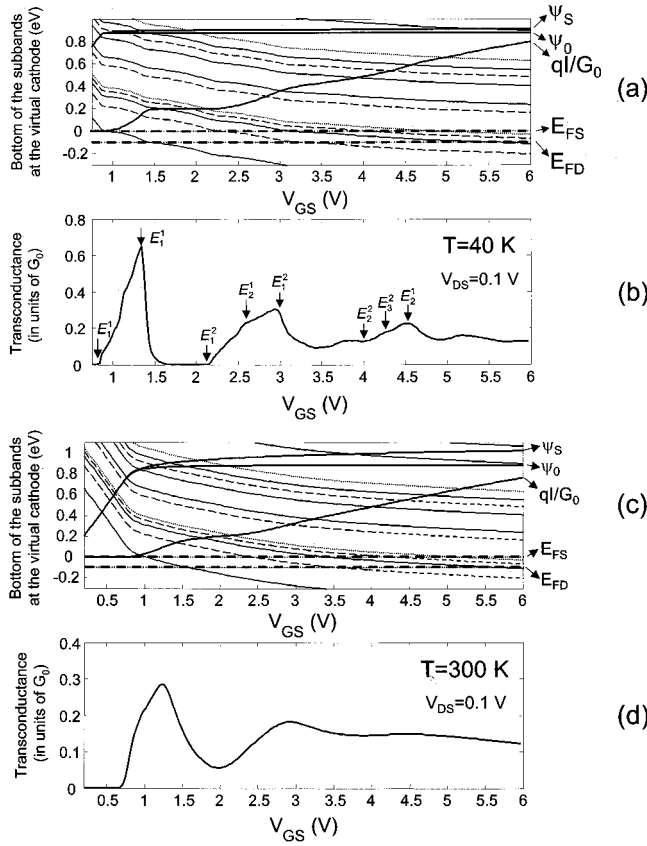


FIG. 4. Bottom of the subbands at the virtual cathode versus the gate voltage at $T=40$ K (a) and $T=300$ K (c). The thin solid, dashed, and dotted lines represent the subbands from the first, second, and third sets of valleys, respectively. In (b) and (d) we show the simulated transconductance at $T=40$ K and $T=300$ K, respectively.

Inserting the one-dimensional (1D) density of states, given by Eq. (A2), into Eq. (11) and after some algebraic manipulations we arrive to

$$Q = \frac{q\sqrt{2kT}}{2W\pi\hbar} \sum_v \sum_n g_v \sqrt{m_d^v} \left\{ \mathcal{J}_{-1/2} \left(\frac{E_{FS} - E_n^v(x_{\max})}{kT} \right) + \mathcal{J}_{-1/2} \left(\frac{E_{FD} - E_n^v(x_{\max})}{kT} \right) \right\}, \quad (12)$$

where

$$\mathcal{J}_{-1/2}(\eta) = \int_0^\infty \frac{x^{-1/2}}{1 + e^{x-\eta}} dx, \quad (13)$$

is the Fermi integral of order $-1/2$, that can be numerically computed by the Blakemore's method.¹⁶ Equation (12) can be conveniently rewritten as

$$Q = \frac{q\sqrt{2kT}}{2W\pi\hbar} \sum_v \sum_n g_v \sqrt{m_d^v} \times \left\{ \mathcal{J}_{-1/2} \left(\frac{E_{FS} - E_1(x_{\max}) - (\Delta E_n^v - \Delta E_1)}{kT} \right) + \mathcal{J}_{-1/2} \left(\frac{E_{FD} - E_1(x_{\max}) - (\Delta E_n^v - \Delta E_1)}{kT} \right) \right\}, \quad (14)$$

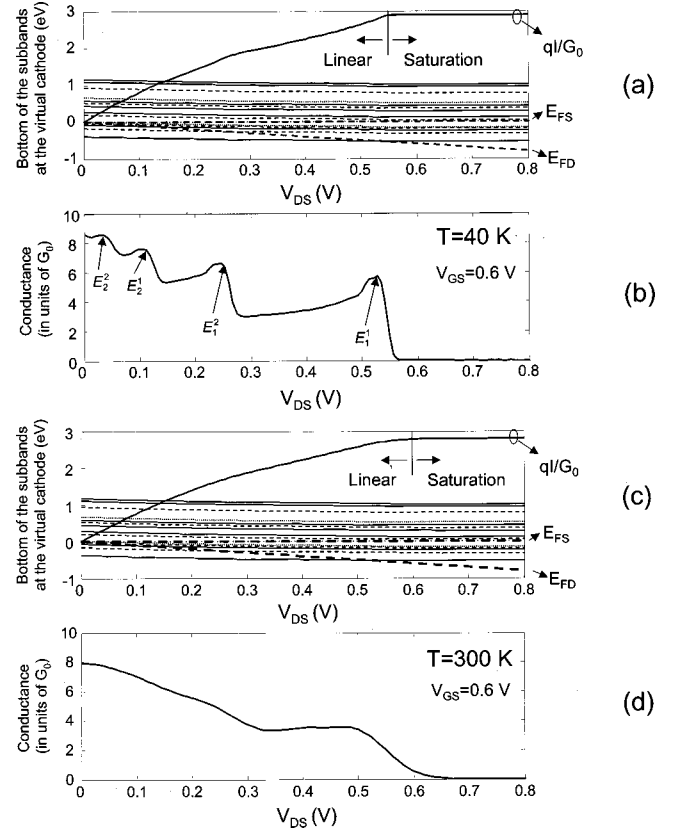


FIG. 5. Bottom of the subbands at the virtual cathode versus the drain voltage at $T=40$ K (a) and $T=300$ K (c). In (b) and (d) we plot the simulated conductance.

where ΔE_n^v is the separation of the bottom of the n th-subband from the bottom of the conduction band, for a two-dimensional infinite square-well potential (see Appendix B); $E_1(x_{\max})$ is the energy of the lowest subband, and ΔE_1 is the separation between $E_1(x_{\max})$ and the bottom of the conduction band. Equation (14) can be numerically solved for the only unknown $E_{FS} - E_1(x_{\max})$ for a given V_{GS} and V_{DS} . We present now an illustrative example using this model.

IV. EXAMPLE

In this section, we show calculations of the transconductance (Fig. 4) and conductance (Fig. 5) of a quantum wire DG-MOSFET at different temperatures. The silicon film thickness and width are 2 and 3 nm, respectively. The oxide thickness is 1.5 nm, and the gate electrodes have a midgap gate work-function (4.61 eV). In Figs. 4(a) and 4(c) we show the bottom of the subbands at the virtual cathode, the surface, and center potentials, and the normalized current (qI/G_0), all of them plotted versus V_{GS} (the Fermi level E_{FS} is taken as a reference). Figures 4(b) and 4(c) show the transconductance at low (40 K) and high temperatures (300 K), respectively. The increase of the gate voltage induces more mobile charge at the virtual cathode, pulling down the maximum of the barrier (respect to the Fermi levels) in order to inject this incremental charge. For the low temperature case, when the gate voltage is below 0.8 V all the subbands are above E_{FS} , and the device operates in the subthreshold region. In this regime, both the current and the transduc-

tance are close to zero. At 0.8 V (the threshold voltage), the first subband (E_1^1) intersects the Fermi level, producing the first peak of the transconductance, and the device enters the above threshold saturation region. The threshold voltage is ≈ 0.7 V at 300 K because the tail of the Fermi function populates subbands above the Fermi level, contributing to the current before the first crossing. By increasing the gate voltage the subbands cross the source and drain Fermi levels producing a structure of peaks and valleys on the transconductance. This is very clear at low temperature, but at room temperature the structure is partially lost because the large extension over the energies of the tail of the Fermi function. In Figs. 5(a) and 5(c) we show the bottom of the subbands at the virtual cathode and the normalized current, plotted versus V_{DS} . Note first of all the slight decrease of the bottom of the subbands with V_{DS} . The mobile charge at the virtual cathode is constant, because V_{GS} is fixed. When V_{DS} is raised, the population of $-k_x$ states at the virtual cathode is gradually reduced. To maintain Q constant, the maximum of the barrier must be pulled down in order to inject more $+k_x$ states from the source, and balance the $-k_x$ states. This effect occurs until the negative states population is totally suppressed at $V_{DS,sat} = 0.55$ V. For low V_{DS} , four subbands are below the two Fermi levels, and the conductance is about $8G_0$ (it would be exactly $8G_0$ at zero Kelvin), each subband contributing with $2G_0$. Increasing V_{DS} , the fourth, third, second, and first subband cross successively E_{FD} , reducing the conductance to about $2G_0$ each one [Fig. 5(b)]. The first subband crossing with E_{FD} determines the onset of the saturation region. At $T = 300$ K, saturation occurs at $V_{DS,sat} = 0.6$ V. The conductance structure is partially lost at this temperature because the tail of the Fermi function smooths the conductance [Fig. 5(d)].

V. COMPARISON TO QUANTUM MECHANICAL SELF-CONSISTENT SIMULATIONS

If the DG-MOSFET has a very large width (W), the confinement in the y -direction is lost, and the electron gas movement is restricted only in the z -direction. In this limit case, the compact model provides good results for the voltage-current characteristics. A width $W = 500$ nm is large enough to remove the confinement in the y -direction. We want to check if the important parameters like the threshold voltage, subthreshold swing, off-current, and on-current are well predicted by our model. For this purpose we have performed a simulation using a DG-MOSFET, with a channel length of 40 nm, a silicon film thickness of 1.5 and 3 nm, an oxide thickness of 1.5 nm, and gate electrodes having a work function of 4.25 eV. The device was simulated at low (100 K) and room temperature (300 K). The results of our model are compared with accurate self-consistent simulations provided by the nanoMOS simulator,^{8,9} based on the nonequilibrium Green's functions.^{17,18} This simulator deals with quantum well DG-MOSFETs, and it can consider up to six two-dimensional subbands. It can be seen from Fig. 6(a) that the subthreshold region is well modeled, which exhibits the ideal subthreshold swing of 60 mV/decade at room temperature (at 100 K this is ≈ 20 mV/decade). Below the threshold voltage

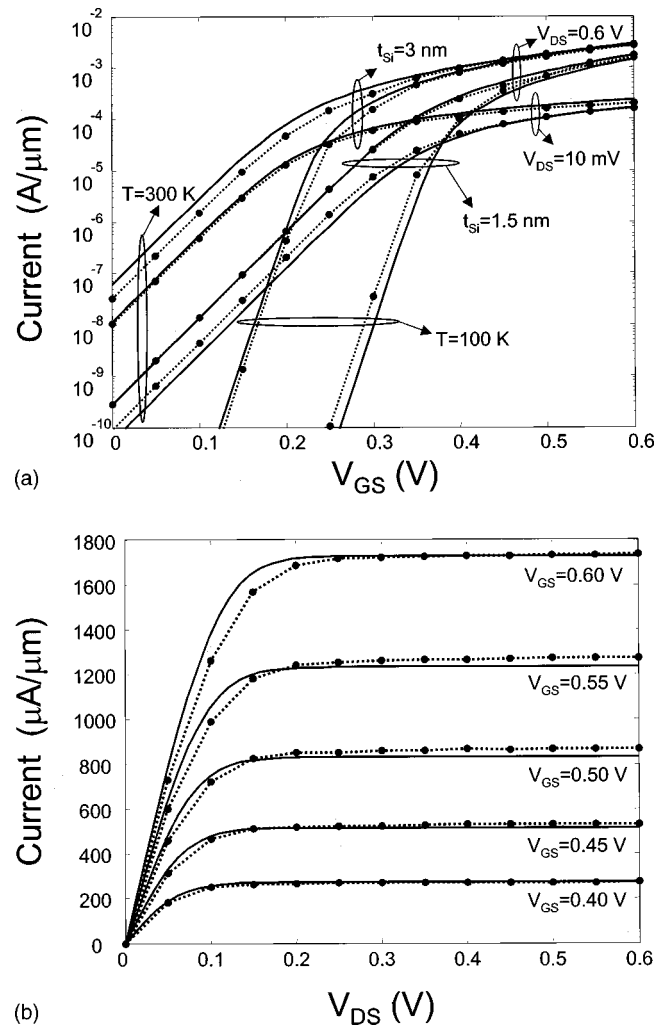


FIG. 6. (a) Transfer characteristic of a quantum well DG-MOSFET of 1.5 and 3 nm silicon film thickness working at 100 and 300 K; (b) output characteristic at room temperature of the quantum well DG-MOSFET with a silicon film thickness of 1.5 nm. In the dotted line, we plot the nanoMOS simulations; in the solid line, the results from the compact model.

the subbands are all above the Fermi level, but they are populated because the tail of the Fermi functions extend over them (Fig. 3). The difference between the transmitted tails of the source and drain Fermi functions explains, in the Landauer approach, the diffusion current, which is the dominant current in the subthreshold region. The linear extrapolated threshold voltage (at low drain bias and $T = 300$ K), extracted from a linear $I - V_{GS}$ plot is about 330 mV for $t_{Si} = 1.5$ nm and 195 mV for $t_{Si} = 3$ nm. It is well predicted by our model even at very small silicon film thickness, where deviation from classical models is very important. The nanoMOS results are 320 and 185 mV, respectively. The off-current and on-current are also well captured by our model [Fig. 6(a)]. For a detailed inspection of the on-current we have plotted in Fig. 6(b) the $I - V_{DS}$ characteristic for a DG-MOSFET with a silicon film thickness of 1.5 nm working at room temperature. The departure on the characteristics predicted by the compact model with respect to the calculated ones by the nanoMOS simulator arises mainly from the number of subbands considered in both the compact model and

nanoMOS simulator, the former dealing with one-dimensional subbands and the latter with two-dimensional subbands. For the nanoMOS calculations we have considered the maximum number of allowed subbands by the program; in the compact model a number exceeding one hundred.

VI. COMPACT MODELING OF CYLINDRICAL GAA-MOSFET AND COMPARISON TO EXPERIMENT

In this section we extend the DG-MOSFET compact model formulated in the above sections to the cylindrical GAA-MOSFET. The following modifications have to be incorporated: (i) the intrinsic carrier concentration, given by Eq. (A3), has to be divided by the area of the circular section ($\pi t_{\text{Si}}^2/4$) instead of the area of the rectangular section (Wt_{Si}); (ii) the separation between the bottom of the conduction band and the bottom of the subbands is given by the theory of the cylindrical quantum potential well (see Appendix B); (iii) the new oxide capacitance (per unit area) for the cylindrical geometry is given by

$$C_{\text{ox}} = \epsilon_{\text{ox}} / \left(\frac{t_{\text{Si}}}{2} \ln(1 + 2t_{\text{ox}}/t_{\text{Si}}) \right), \quad (15)$$

(iv) the potential in the channel is a solution of the Poisson's equation for the cylindrical geometry (see Appendix C). The surface and center potentials appearing in Eqs. (7) and (9) must be modified in accordance with it; and (v) for the cylindrical geometry, there are two sets of valleys, the first one is four-fold degenerated with a density of states effective mass $m_d^1 = m_T$, and the second set, two-fold degenerated, has a density of states effective mass $m_d^2 = m_L$.

In order to test our approach we have compared the results given by our compact model with an experiment reported in the literature.¹⁹ The GAA-MOSFET of the experiment has a diameter (t_{Si}) of 65 nm, the insulator is silicon oxide with a thickness of 35 nm. The work-function of the gate material has been fixed to 4.43 eV in the compact model to match the experimental threshold voltage (0.48 V). To observe one-dimensional subband effects for this size, the operation temperature was 70 mK. In Fig. 7 we compare the measured transconductance (dotted line) with the simulated results by the compact model (solid line). The position of the five main peaks is well predicted by the compact model. A detailed analysis reveals eight subbands below E_{FS} in the analyzed voltage range. When the gate voltage is large enough the subband bottom energy reaches E_{FD} . Every time one subband crosses E_{FS} (E_{FD}) a positive (negative) peak on the transconductance is recorded. The transconductance step is given by the derivative of the subband bottom energy with respect to the gate voltage multiplied by $4G_0$ or $2G_0$, depending whether the subband belongs to the first or second set of valleys, respectively.

VII. CONCLUSIONS

The need for compact models of nanoscale transistors is increasingly important for future development of computer-aided-design tools at the circuit level. In this work we have presented a physics-based model for the ballistic quantum

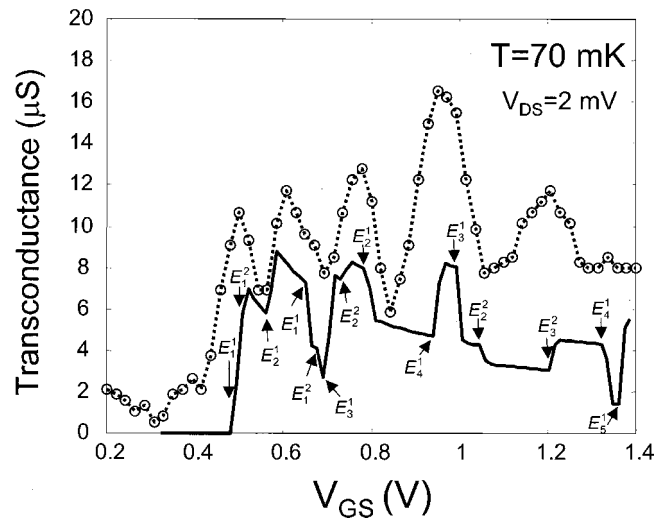


FIG. 7. Comparison between experimental (dotted line) and simulated (solid line) transconductance of a cylindrical GAA-MOSFET of 65 nm width.

wire and quantum well MOSFET, derived from the Landauer transmission theory. The main novelty is that it extends modeling concepts of quantum well MOSFETs, to structures of lower dimensionality (quantum wire), and is easily adaptable to gate structures as the single-gate, double-gate, or gate-all-around. The proposed model works in all the operation regions, below and above threshold, for low and high temperatures, incorporating effects of multisubband conduction, and taking into account the band structure of silicon. To verify the model we have proposed numerical experiments comparing the results with self-consistent simulations. The quantum wire model was used to predict the transconductance structure of a real device, showing good agreement.

APPENDIX A: INTRINSIC CARRIER DENSITY OF A QUANTUM WIRE

The intrinsic carrier density is evaluated by integrating the product $N_{1D}(E) \cdot f(E)$ over all the energies in the conduction band and summing for all the subbands and valleys

$$n_i = \sum_v \sum_n g_v \int_{E_n^v}^{\infty} N_{1D}^v(E) f(E) dE. \quad (A1)$$

$N_{1D}^v(E)$ is the one-dimensional density of states (with the spin degeneracy included) given by

$$N_{1D}^v(E) = \frac{\sqrt{2m_d^v}}{\pi\hbar} \sum_n \frac{H(E - E_n^v)}{\sqrt{E - E_n^v}}, \quad (A2)$$

where the density of states effective mass (m_d^v) is given by $m_d^1 = m_T$ for the first set of valleys, $m_d^2 = m_T$ for the second set, and $m_d^3 = m_L$ for the third set; $f(E)$ denotes the Fermi function with a Fermi level (E_{Fi}) located at the middle of the silicon band gap [Fig. 2(b)]. Replacing Eq. (A2) into Eq. (A1) and integrating, one obtains

$$n_i = \frac{\sqrt{2kT}}{\pi\hbar} \sum_v \sum_n g_v \sqrt{m_d^v} \mathcal{J}_{-1/2} \left(\frac{E_{\text{Fi}} - E_n^v}{kT} \right). \quad (A3)$$

Note that $E_{\text{Fi}} - E_n^v = -E_g/2q - \Delta E_n^v$, where ΔE_n^v is the difference between the bottom of the n th-subband (E_n^v) and the bottom of the conduction band (see Appendix B). The units in Eq. (A3) are in m^{-1} because we have used a one-dimensional density of states. To match with the dimensions required in the Poisson's equation (m^{-3}), the intrinsic carrier density must be divided by the two confining dimensions W and t_{Si} . Once the intrinsic carrier density is known we can determine the sheet density of mobile charge through Eqs. (7) and (9).

APPENDIX B: CONFINEMENT IN A TWO-DIMENSIONAL INFINITE SQUARE-WELL AND CYLINDRICAL-WELL POTENTIALS

For the DG structure, we can assume that the electrons are confined in a two-dimensional infinite square-well potential. This assumption ignores the exact form of the potential into the channel that depends on V_{GS} , being more valid in the subthreshold regime because the potential channel is approximately flat. However this hypothesis captures the essential physics. For the two-dimensional infinite square-well, the separation between the bottom of the conduction band and the bottom of the subbands is given by

$$\Delta E_{n_y, n_z}^v = \frac{\hbar^2}{2} \left\{ \frac{(\pi n_y)^2}{m_y^2 W^2} + \frac{(\pi n_z)^2}{m_z^2 t_{\text{Si}}^2} \right\}, \quad (\text{B1})$$

where $(m_y^1, m_z^1) = (m_T, m_L)$; $(m_y^2, m_z^2) = (m_L, m_T)$; and $(m_y^3, m_z^3) = (m_T, m_T)$.

For the cylindrical GAA structure, the electrons are confined in a quantum cylindrical-well, for which the separation between the bottom of the conduction band and the bottom of the subbands is given by²⁰

$$\Delta E_{n_r, n_\alpha}^v \approx \frac{2\hbar^2}{m_\perp^2 t_{\text{Si}}^2} \left\{ \left(n_r + \frac{1}{2} |n_\alpha| - \frac{1}{4} \right) \pi \right\}^2, \quad (\text{B2})$$

where the effective mass appearing in Eq. (B2) is $m_\perp^1 = \sqrt{m_L m_T}$ for the first set of valleys (four-fold degenerated) and $m_\perp^2 = m_T$ for the second set of valleys (two-fold degenerated).²¹ For the sake of simplicity, the three-integer subindex notation (v, n_y, n_z) , (v, n_r, n_α) , identifying the valley and the subband is replaced by a two-integer notation in the main text; i.e., $\Delta E_{n_y, n_z}^v$, $\Delta E_{n_r, n_\alpha}^v \rightarrow \Delta E_n^v$.

APPENDIX C: ELECTROSTATICS OF A CYLINDRICAL GAA-MOSFET

The potential for a cylindrical silicon channel surrounded by an insulator and a gate electrode has been studied by Auth and Plummer.²² In that work, a doped channel is assumed. The charge term in the Poisson's equation is constant (dominated by the dopant concentration), and a parabolic solution of the type $\psi(r) = \psi_0 + Cr^2$ can be derived, being $\psi(r)$ the potential at a distance r from the center of the quantum wire, ψ_0 the potential at the center, and C a constant to be determined. Our case is more difficult to solve exactly

because the channel is intrinsic silicon, being the charge term in the Poisson's equation proportional to $\exp[q\psi(r)/kT]$. However, we can consider the above solution for the subthreshold and moderate inversion because the parabola shows very weak dependence with the radius, and the charge term can be approximated by $\exp(q\psi_0/kT)$. In this situation, the following equation is obtained if we substitute the proposed parabolic solution into the Poisson's equation:

$$\frac{8\epsilon_{\text{ox}}(V_{\text{GS}} - \Delta\phi - \psi_0)}{\epsilon_{\text{Si}} t_{\text{Si}}^2 \ln \left(1 + 2 \frac{t_{\text{ox}}}{t_{\text{Si}}} \right) + \frac{\epsilon_{\text{ox}} t_{\text{Si}}^2}{2}} \approx \frac{q}{\epsilon_{\text{Si}}} e^{q\psi_0/kT}. \quad (\text{C1})$$

This equation can be solved for ψ_0 for a given V_{GS} . With this parameter, the surface potential ψ_s entering in Eq. (9), can be obtained from

$$\psi_s = \frac{2 \ln(1 + 2 t_{\text{ox}}/t_{\text{Si}}) \psi_0 + \epsilon_{\text{ox}}(V_{\text{GS}} - \Delta\phi)/\epsilon_{\text{Si}}}{2 \ln(1 + 2 t_{\text{ox}}/t_{\text{Si}}) + \epsilon_{\text{ox}}/\epsilon_{\text{Si}}}. \quad (\text{C2})$$

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